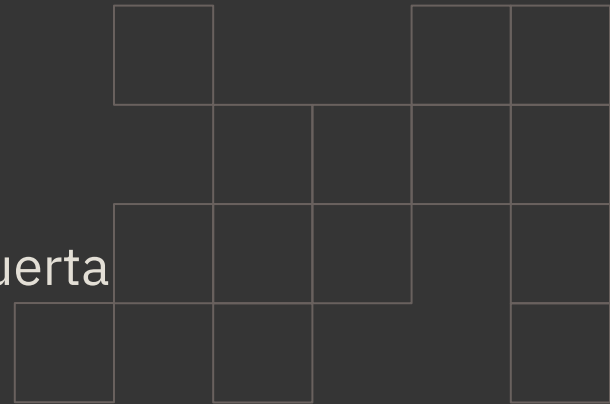




AirTrak

By Abdul Hossain and Zuriel Huerta

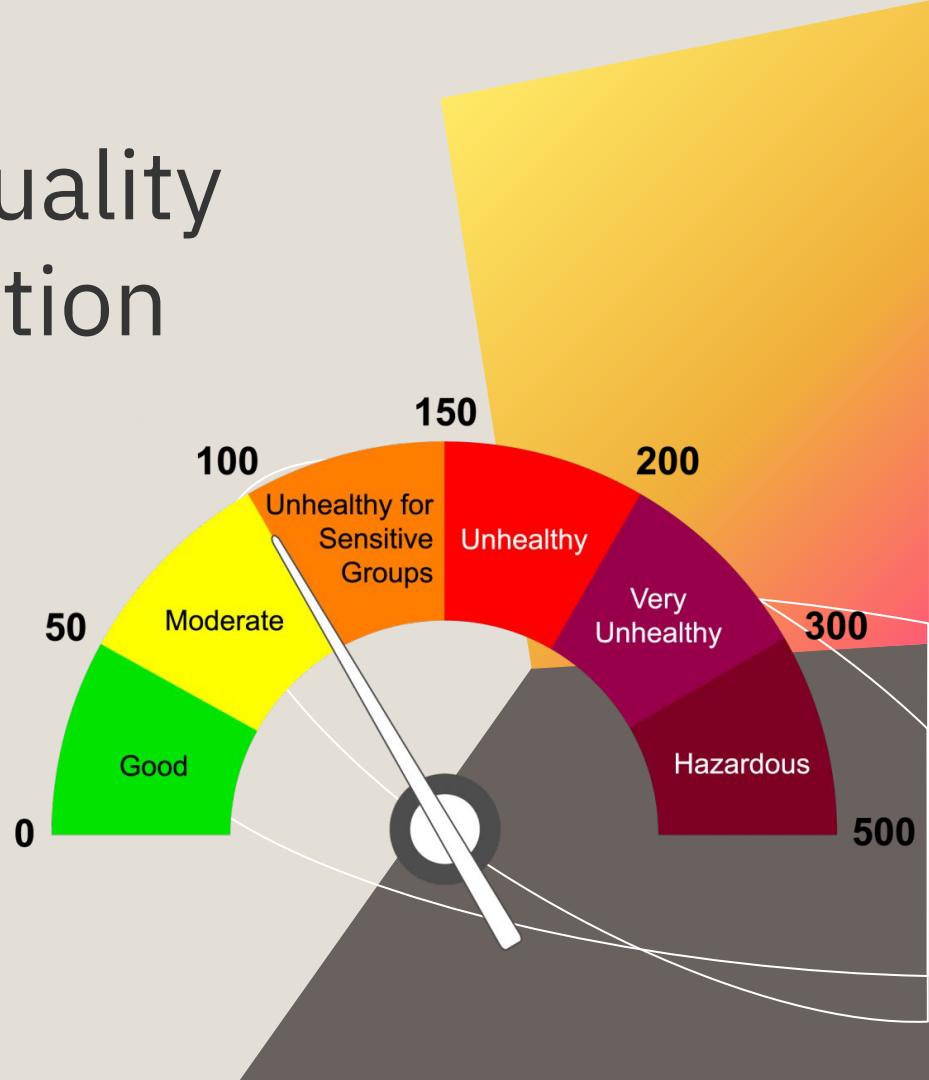
<https://github.com/abdu1h-dev/AirTrak>



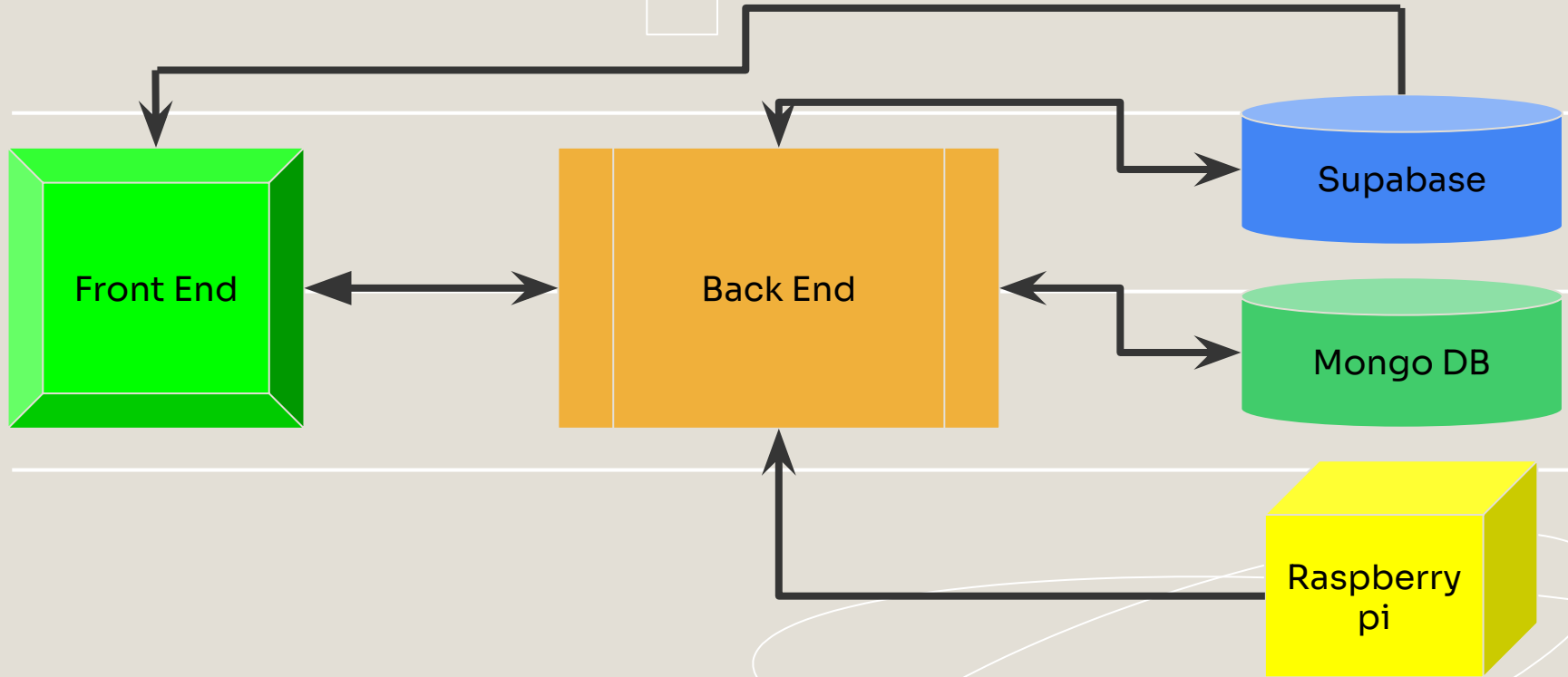
Access real-time air quality data via a web application

Air quality is a very important aspect of comfortable living and daily maintenance.

Our app allows you to place sensors around you chosen area and track these finding in real time. Air quality on its own is a very complex thing to track consistently in non-commercial settings, but with our app we can track real time finding, and concentrate them down into easy and decisive metrics.



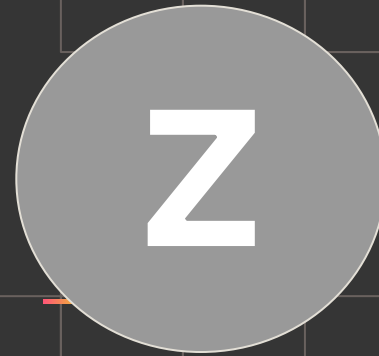
(B) Project Architecture





Abdul Hossain
BackEnd

I worked largely on Backend Development and decided on fastapi as the framework. I had some experience with database design and hosting them



Zuriel Huerta
FrontEnd

I worked on the Frontend for this project. I have some experience with but only at a beginner level, and was exposed to new aspect of javascript and frontend design

Front End Technologies

Charting

Creating real time graphs that express the ever changing data in a easily approachable and understandable manner

Framework

Using a framework and its quirks to give a seamless experience that doesn't cause too much buffering..

API Calls

The Front End for our project relies on different API calls from different services each with different syntax formats and outputs. These api calls (when used properly are faster than the time it take to instantiate a websockets connection)

Back End Technologies

1

Fast Api

Fast API is a straightforward backend web framework that we hosted on render using a supporting library called uvicorn

2

Websockets

A computer communications protocol that facilitates two-way, real-time communication over a single, persistent TCP connection

3

Database Design

Databases used to store Air quality data and another to store usernames and passwords

Challenges Faced And Lessons Learned

1

Parallelizing Work

Figuring out way to work together on the same task in a way that was productive as opposed to racing towards and answer

2

Database Design

Constant complication in data cleaning, structuring, insertion, and pulling caused us to re-evaluate and remake the database schemas multiple times

3

Hosting

Hosting and figuring out the syntax of each hosting platform was difficult and so was rapid iteration and testing when hosting the backend and front you are testing

4

Raspberry Pi

Raspberry pi documentation in the context of fastapi is scarce and its non-persistent connection led us to change the technologies used in our project

Next steps

Password Encryption

Password Encryption would be a useful and important feature. There are useful functions in the MongoDB documentation for this purpose but requires us to restructure the basic username and password database.

Location Tracking and Real Time Air Quality mapping

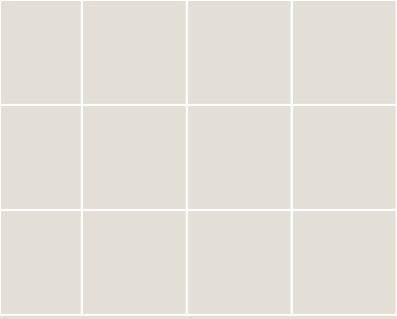
Public Data set are expansive and can be used in correlation with private air quality data to create something like an Air Quality map. But implementing location tracking on the sensors would be very labour intensive.

Sharing private air quality data Publicly

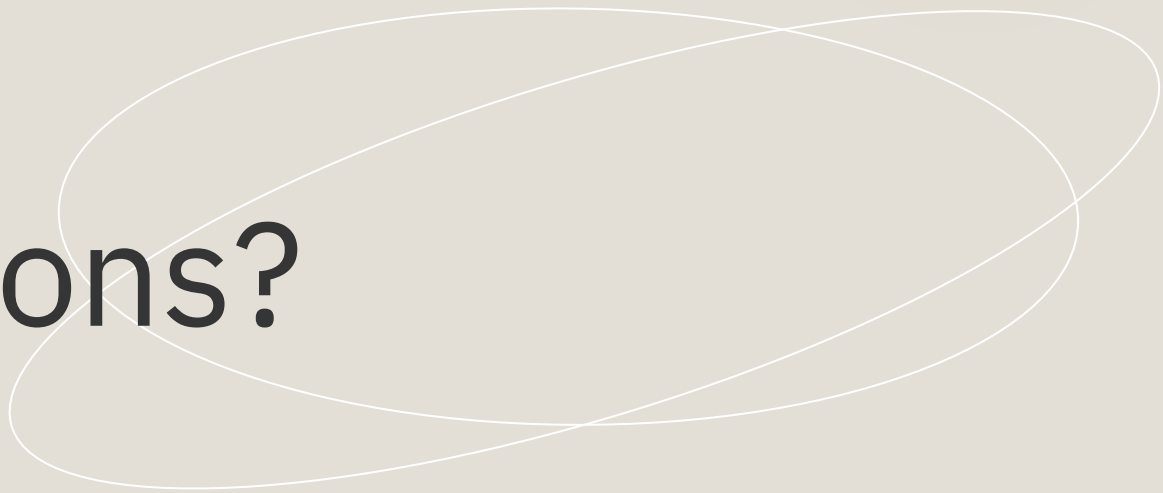
Doing this would require us to restructure the Supabase database that we set up to be much more modular and this modularity would require a more dynamic backend to cope with when inserting and pulling data



Thank you



Any questions?
Ask away!



Github Link :

<https://github.com/abdu1h-dev/AirTrak>